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High Altitude Problems, Research Findings and Recommendations

HRRP Meeting

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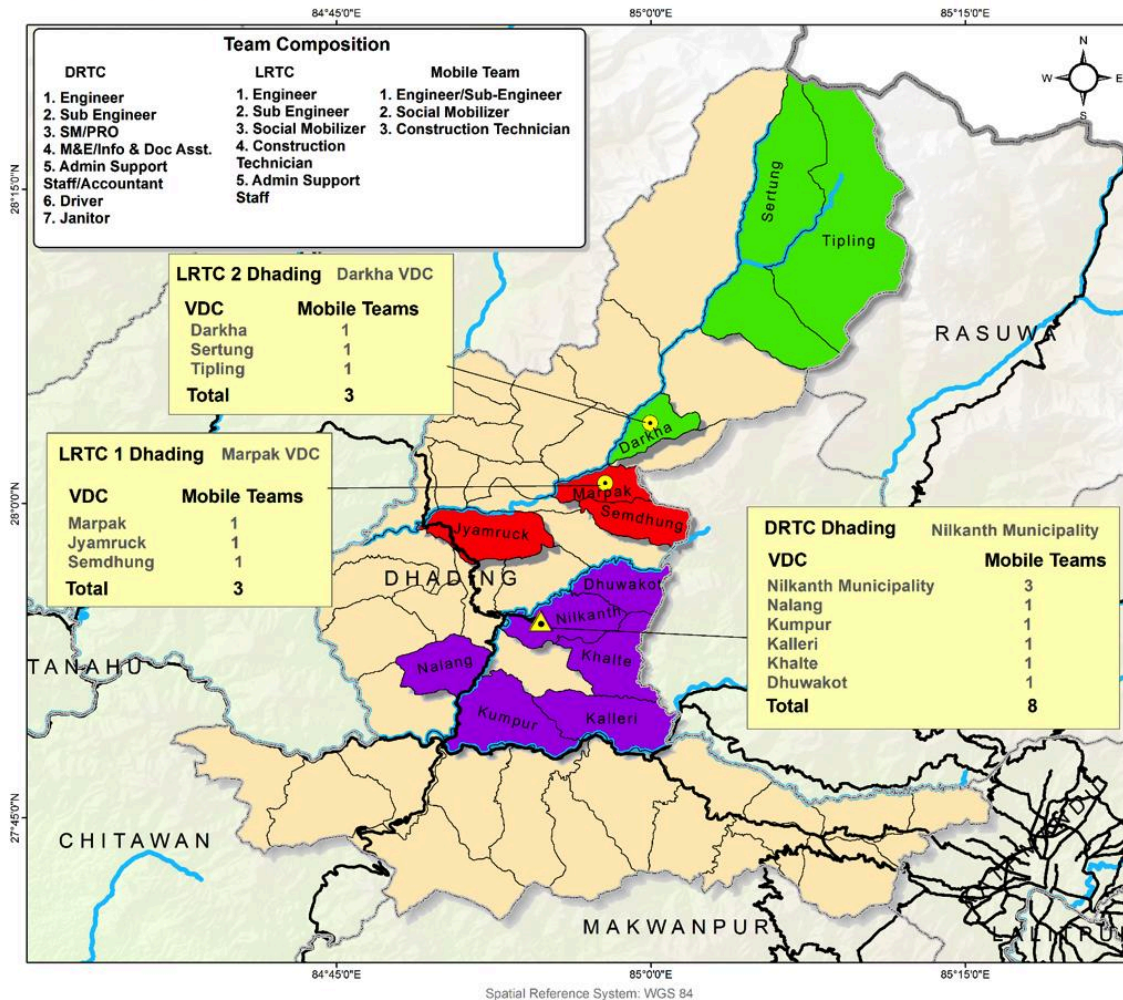


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Basis: Work in Northern Area

BALIYO GHAR PROGRAM



RESOURCE CENTER Dhading



10 5 0 10
KM

Abbreviations

DRTC: District Resource and Training Center
LRTC: Local Resource and Training Center
MT: Mobile Team
SM: Social Mobilizer
PRO: Public Relation Officer

Data Source: NGIIP, Government of Nepal

Legend

- ▲ DRTC Headquarter
- LRTC Headquarter
- International Boundary
- District Boundary
- VDC/Municipality Boundary
- Major Roads
- Major Rivers
- Program District

Resource Center Area

- Dhading DRTC
- Dhading LRTC 1
- Dhading LRTC 2

TIPLING





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SHERTUNG



Existing Building typologies

- Dry Stone masonry or Stone in mud buildings with heavy Stone roofs
- Presence of corner stones



Baliyo Ghar Program





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Existing Building typologies

- No other Earthquake Resisting Element





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New Building typologies

- Dry Stone masonry with light CGI sheets
- No earthquake resisting elements
- Presence of corner stones
- Use Gable wall



New Construction after Earthquake



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New Building typologies

- Timber walls at the front part and inner side of all stone walls



Many Buildings!

New Building typologies

- No proper connection of ridge beam, rafter and bottom chord
- No proper connection of roof elements with wall





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High Altitude Area Technical Problems



- RCC reinforcing elements cannot be used as it is difficult to transport cement and steel since these sites are very remote and have no road access
- Only very soft timbers (**Gobre Salla**) are available as reinforcing elements to be used in bands, vertical elements and stitches

High Altitude Area Technical Problems

“We cannot use timbers as band and corner stiches” – locals

- Only Gobre Salla, soft pine wood available
- It's only being used for roofs
- After 4/5 years of roof covering, the timber decays
- People use these timbers planks to support all stone walls from with the assumption that, during earthquake these timber will stop stone wall falling inside the house, and the stone wall will fall outside, so no casualties would happen

Findings # 1

**Dry Stone Wall Perform
Better than Stone in Mud**

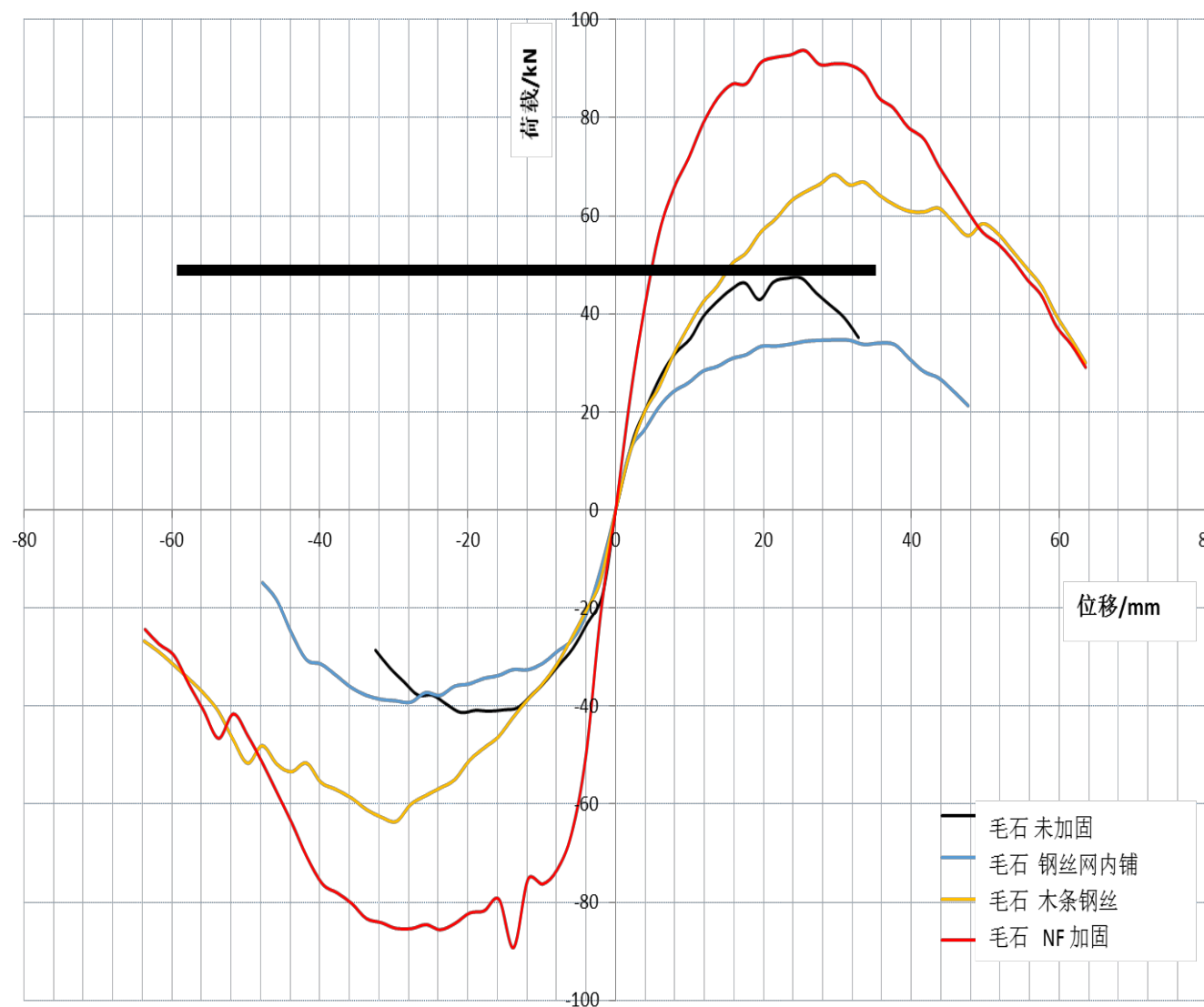
Progress: Test #1: Round Stone wall test



Movement of the wall at a ultimate displacement of 32mm

No Direct Budget under Baliyo Ghar : Work with Other Universities worldwide

Comparison of loading-displacement curves of round stone wall specimens



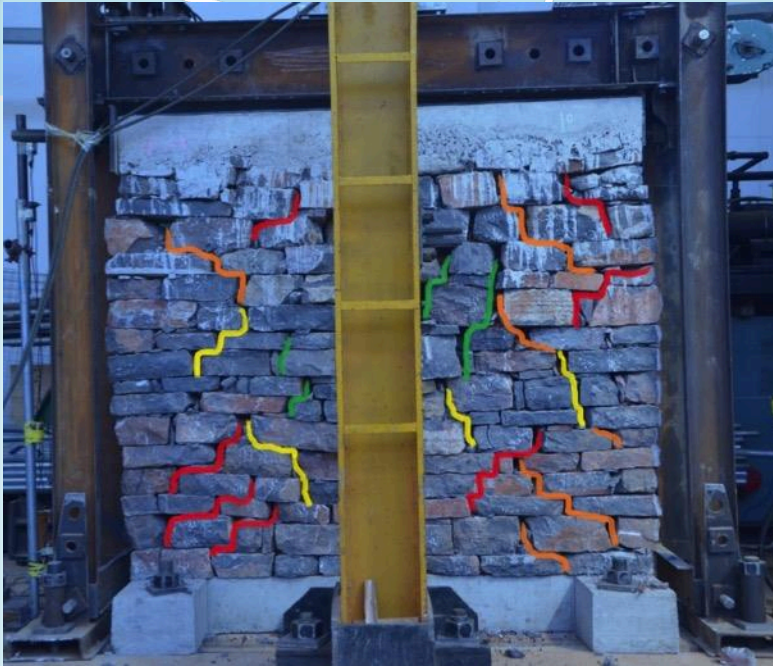
Black: Unreinforced
Yellow: Wooden band&post + gabion wire
Red: Tarpaulin strips
Blue: Wooden post + gabion wire



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S1: Non-retrofitted (NR) as a reference specimen



Green :

18m

m

Yellow :

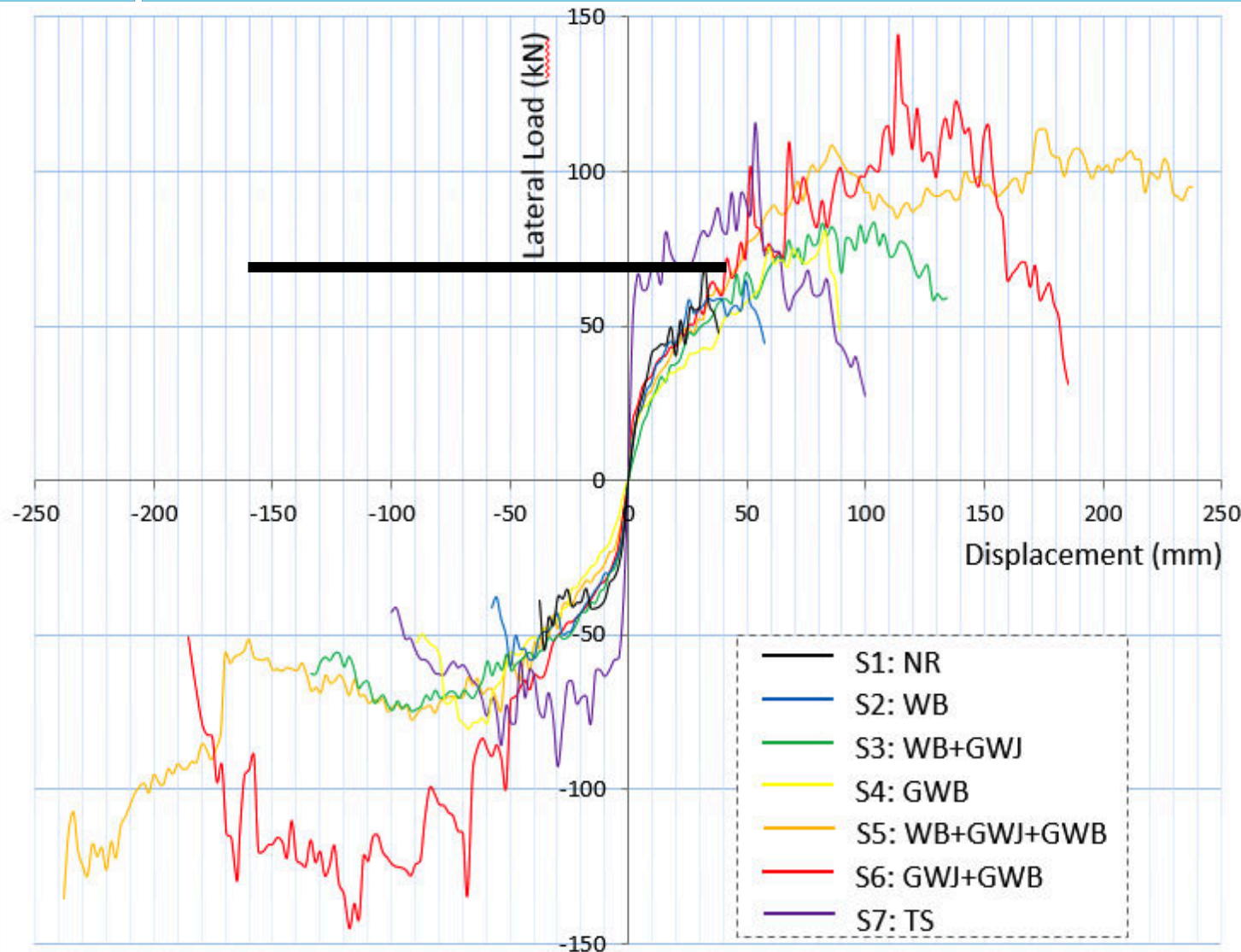
24m

m

Orange :

28m

Comparison of loading-displacement curves of flat stone wall specimens



Findings # 2

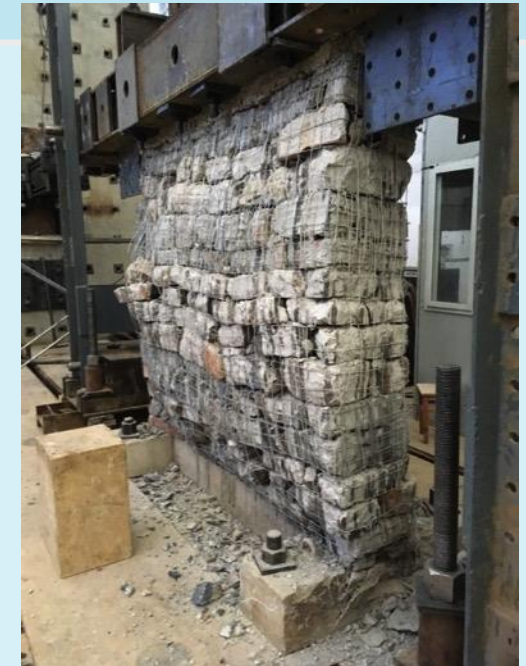
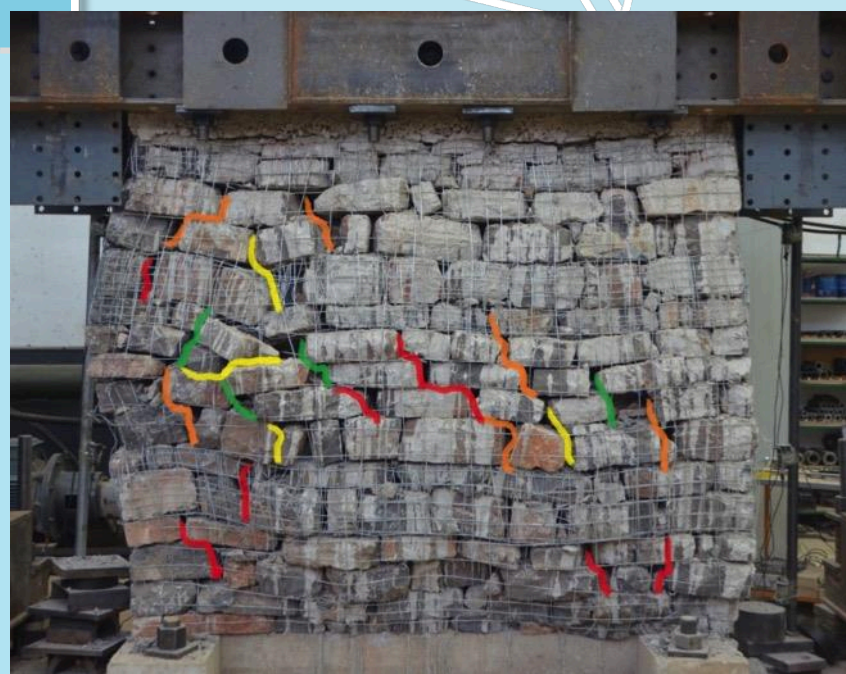
**Welded Gabion Mesh or
Geo-Grid Mesh can be
alternate to Soft-Timber**



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S6: Gabion wire jacketing and bandages (GWJ+GWB)



Green:

110

mm

Yellow:

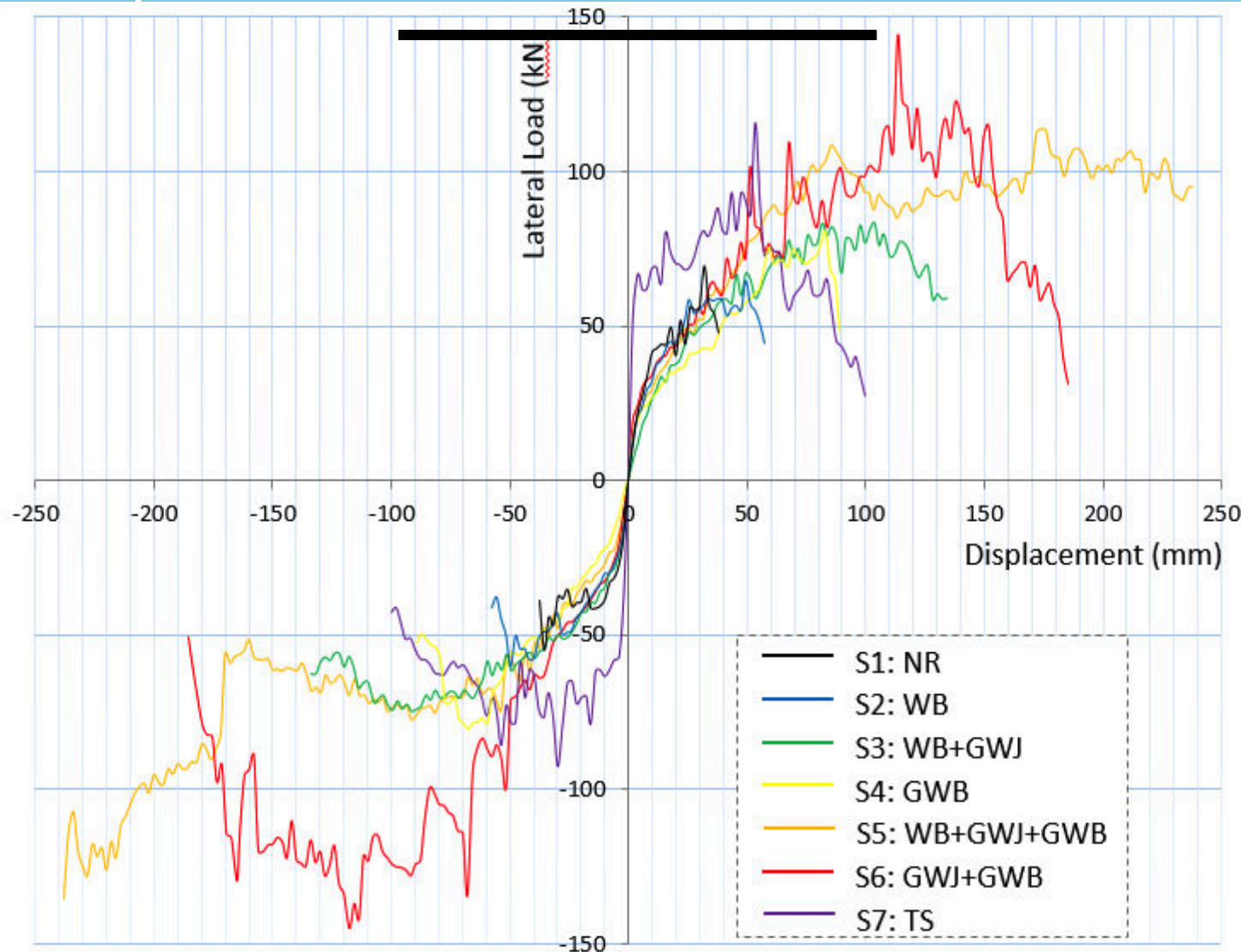
124

mm

Orange:

150

Comparison of loading-displacement curves of flat stone wall specimens



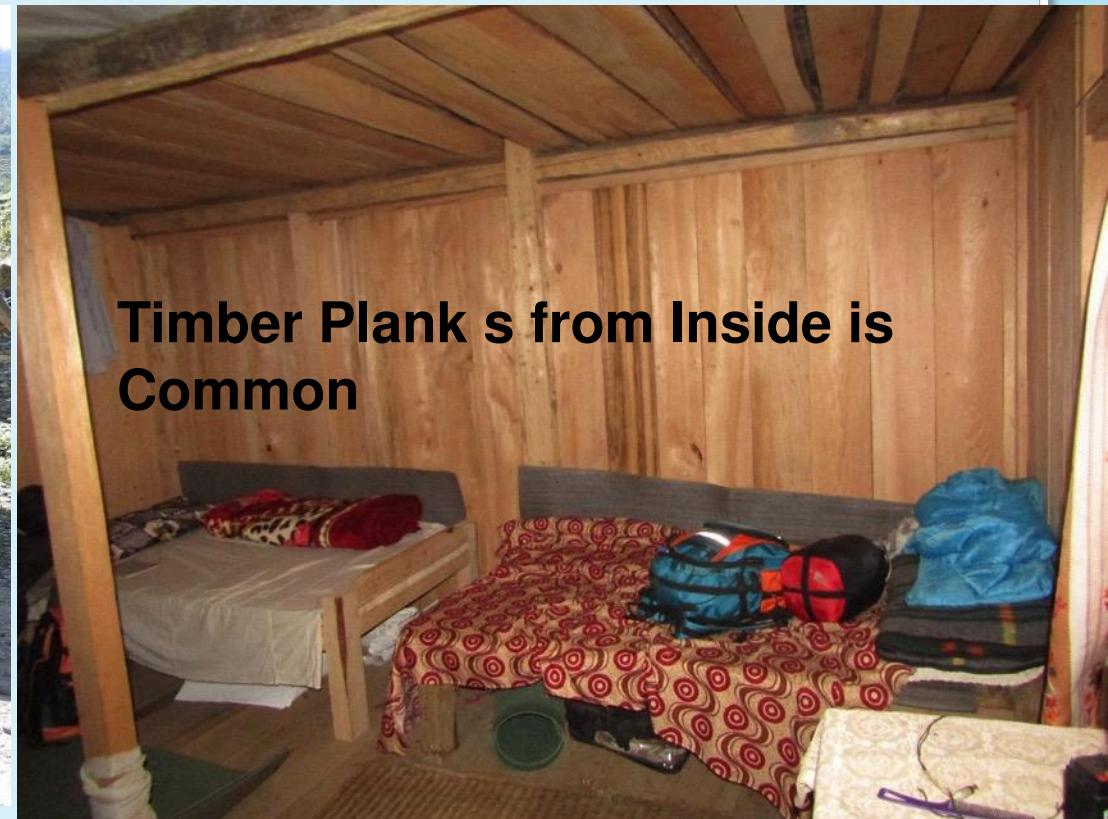
Findings # 3

Quantity vs Quality of Timber

Community Agree to Use more Timber

New Building typologies

- Timber walls at the front part and inner side of all stone walls



Timber Plank s from Inside is Common

Need Technology that Support their Way of Life and Functioning!



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Some recommendations



- To increase the longevity of those soft timbers, protect the timber by seasoning, drying or using bitumen based paint
- Even then, in 10 or 15 years period time, the timber will get decayed. So, the other idea is to replace the timbers after it gets decayed
- Timber can be used as timber frame in such a way that they can replace it periodically



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Some recommendations



- Hence instead of placing timber inside the wall in bands and vertical elements, it can be placed on outer faces of wall on both side and then connect them with gabion wires or steel bolts
- But it requires wider technical discussions to finalize
- This new system should be included in policy documents or checklist also
- The connection of roof timber elements with each other and with the wall also needs to be improved



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Some recommendations



- Gabion Welded wire mesh and/or Geo-Grid Mesh are some of the alternatives

**Whatever we decide, Let's decide it Quickly!
Many Houses are already under Construction!**